

Detailed Contents

1 Evolution, the Themes of Biology, and Scientific Inquiry 2

CONCEPT 1.1 The study of life reveals unifying themes 3

- Theme: New Properties Emerge at Successive Levels of Biological Organization 4
- Theme: Life's Processes Involve the Expression and Transmission of Genetic Information 6
- Theme: Life Requires the Transfer and Transformation of Energy and Matter 9
- Theme: From Molecules to Ecosystems, Interactions Are Important in Biological Systems 9

CONCEPT 1.2 The Core Theme: Evolution accounts for the unity and diversity of life 11

- Classifying the Diversity of Life 12
- Charles Darwin and the Theory of Natural Selection 14
- The Tree of Life 15

CONCEPT 1.3 In studying nature, scientists form and test hypotheses 16

- Exploration and Observation 17
- Gathering and Analyzing Data 17
- Forming and Testing Hypotheses 17
- The Flexibility of the Scientific Process 18
- A Case Study in Scientific Inquiry: Investigating Coat Coloration in Mouse Populations* 20
- Variables and Controls in Experiments 20
- Theories in Science 21

CONCEPT 1.4 Science benefits from a cooperative approach and diverse viewpoints 22

- Building on the Work of Others 22
- Science, Technology, and Society 23
- The Value of Diverse Viewpoints in Science 24

Unit 1 The Chemistry of Life 27

Interview: Kenneth Olden 27

2 The Chemical Context of Life 28

CONCEPT 2.1 Matter consists of chemical elements in pure form and in combinations called compounds 29

- Elements and Compounds 29
- The Elements of Life 29
- Case Study: Evolution of Tolerance to Toxic Elements* 30

CONCEPT 2.2 An element's properties depend on the structure of its atoms 30

- Subatomic Particles 30
- Atomic Number and Atomic Mass 31
- Isotopes 31
- The Energy Levels of Electrons 32
- Electron Distribution and Chemical Properties 34
- Electron Orbitals 35

CONCEPT 2.3 The formation and function of molecules and ionic compounds depend on chemical bonding between atoms 36

- Covalent Bonds 36
- Ionic Bonds 37
- Weak Chemical Interactions 38
- Molecular Shape and Function 39

CONCEPT 2.4 Chemical reactions make and break chemical bonds 40

3 Water and Life 44

CONCEPT 3.1 Polar covalent bonds in water molecules result in hydrogen bonding 45

CONCEPT 3.2 Four emergent properties of water contribute to Earth's suitability for life 45

- Cohesion of Water Molecules 45
- Moderation of Temperature by Water 46
- Floating of Ice on Liquid Water 48
- Water: The Solvent of Life 49
- Possible Evolution of Life on Other Planets 50

CONCEPT 3.3 Acidic and basic conditions affect living organisms 51

- Acids and Bases 51
- The pH Scale 51
- Buffers 52
- Acidification: A Threat to Our Oceans 53



4 Carbon and the Molecular Diversity of Life 56

CONCEPT 4.1 Organic chemistry is key to the origin of life 57

CONCEPT 4.2 Carbon atoms can form diverse molecules by bonding to four other atoms 58

- The Formation of Bonds with Carbon 58
- Molecular Diversity Arising from Variation in Carbon Skeletons 60

CONCEPT 4.3 A few chemical groups are key to molecular function 62

- The Chemical Groups Most Important in the Processes of Life 62
- ATP: An Important Source of Energy for Cellular Processes 64
- The Chemical Elements of Life: A Review 64